

Module 3 — DADA2 in Galaxy to phyloseq

BIOL 4810

2026-04-02

Module 3 — DADA2 in Galaxy to phyloseq

Goal: Process amplicon sequencing reads in Galaxy using DADA2, examine key outputs, then prepare the files needed for analysis in **phyloseq**.

Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

0) Your dataset

0A. Sample set / project name:

Bird microbiome

0B. Sequencing type:

paired-end Illumina

0C. Metadata grouping variable you plan to analyze in phyloseq:

infection status

DADA2 workflow order

Use the Galaxy tools in this order:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →  
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → sequence  
counts
```

Reminder: The point of this workflow is to turn messy raw reads into a clean ASV table with taxonomy that can be analyzed ecologically in **phyloseq**.

Part 1 — Quality profiles

1A) Forward read quality profile

What is the purpose of this step?

The quality profile tool aids in visualizing and understanding the distribution of respective quality scores.

Upload screenshot: forward plotQualityProfile PDF

Choose File no file selected

Where does quality begin to noticeably decline in the forward reads?

around 220

Chosen forward truncation length:

NA

1B) Reverse read quality profile

What is usually different about reverse-read quality compared with forward-read quality?

Reverse reads are typically lower quality than forward reads because by the time the first read has been sequenced and the second can start, the signal has degraded and accuracy declines.

Upload screenshot: reverse plotQualityProfile PDFChoose File  Screenshot... 21.48.08**226: dada2: plotQualityProfile on dataset 27-46 and collection 47: reverse reads**

^

ok

37.4 KB

format **pdf** database ? size **37.4 KB** Preview Visualize Details Edit**Where does quality begin to noticeably decline in the reverse reads?**

around 200 bp

Chosen reverse truncation length:

NA

1C) Truncation decision**Explain why you chose these truncation values.**

I chose truncation values a margin away from those values whereby quality began to drop in order to keep as much overlap as possible for merging. Willing to tolerate maybe a quality score above 25- up to you.

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim**What filterAndTrim does:**

This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

Trimming from the ends and filtering enhances the accuracy of downstream processing by helping to eliminate sequencing errors.

Parameters used:

"Truncate read length" was set to 220; selected yes to "separate filters and trimmers for reverse reads;" "trim start of each read" was set to 10; "truncate read length" was set to 205.

What could happen if you trim too aggressively?

Poor merging and loss of usable data can result from overtrimming, reducing coverage/depth of sequence(s).

What could happen if you do not trim enough?

Low quality sequences can increase error rates.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim`, your output is a **paired collection**.

Before continuing, you must split this into **forward and reverse collections**.

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads.
If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:

This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants from sequencing mistakes.

Why is this step important?

The `learnErrors` step is important because it determines how and how often errors occur.

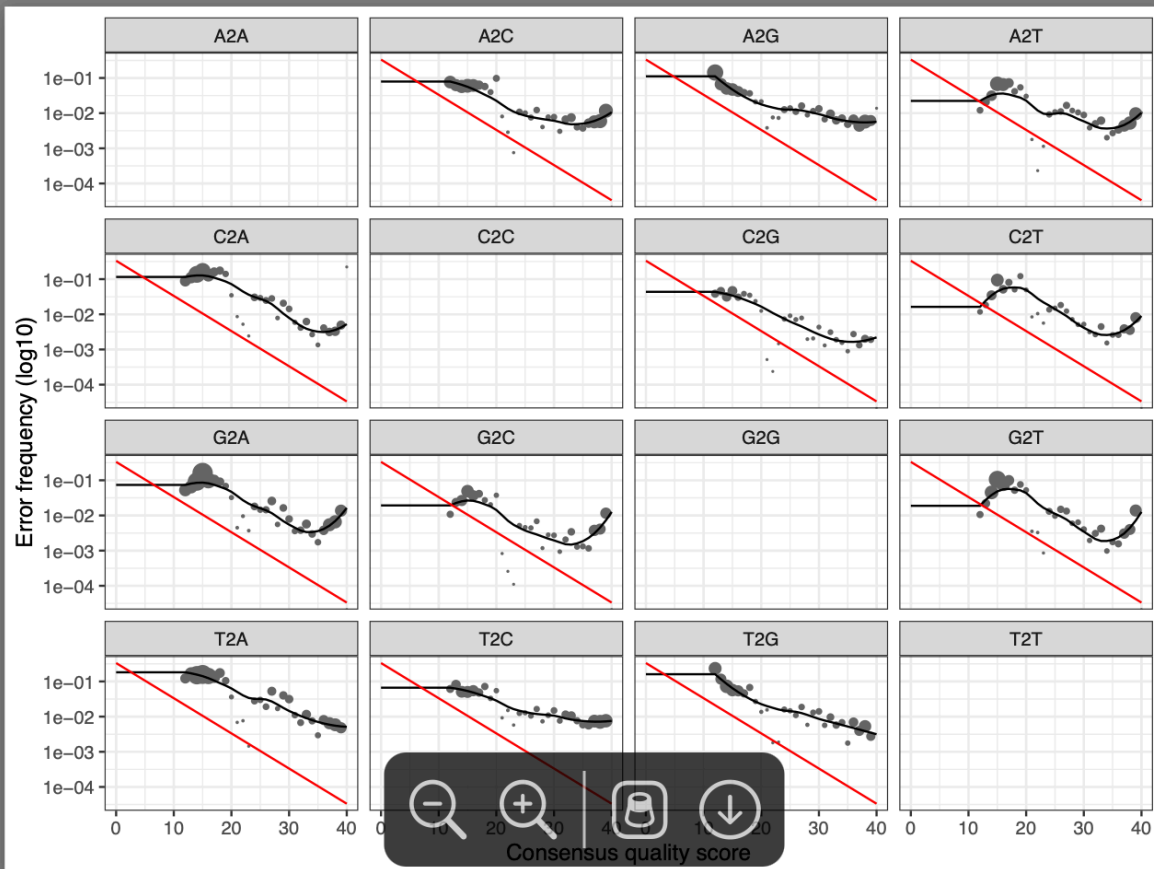
//

Upload screenshot: learnErrors PDFChoose File  Screenshot... 22.21.22**183: dada2: learnErrors collection 160: error rates plot
forward**

^

ok

31.7 KB

format **pdf** database ? size **31.7 KB** Preview Visualize Details Edit**Does the fitted model generally follow the observed error pattern?**

The fitted model mostly follows the observed error pattern.

Why is learning error rates important before denoising?

Learning error rates is important to perform before denoising because it allows dada2 to distinguish between true sequence variation and sequencing errors.

Part 4 — dada (denoising)

What dada does:

This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

When provided filtered reads and error reads, it uses the learned error model to correct errors and reconstruct true sequences (amplicon sequence variants).

How is ASV inference different from OTU clustering?

OTU clustering groups of sequences into taxa based on similarity threshold without modeling errors, whereas ASV Inference corrects sequencing errors at the nucleotide level. ASV provides more precision in VL's opinion.

Part 5 — mergePairs

What mergePairs does:

This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

Forward and reverse reads are being merged to assess confidence of sequence accuracy.

Why do forward and reverse reads need to overlap?

Overlap between forward and reverse reads confirms that they match and indicates accuracy.

If reads fail to merge, what is one likely reason?

Reads may fail to merge if there isn't enough overlap.

Part 6 — makeSequenceTable

What makeSequenceTable does:

This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

The ASV table represents diversity with precise comparison of sampled sequences. there are about 1600 ASVs.

What are the rows, columns, and cell values in the sequence table?

Columns are ASVs, samples, and frequency of ASV appearance in each sample.

Part 7 — removeBimeraDenovo

What removeBimeraDenovo does:

This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

Chimeras are sequences artificially generated during PCR and because they don't actually exist in the original sample the presence of chimeras compromises sequence accuracy and downstream processing.

Why are chimeras a problem for interpretation?

The presence of chimeras can lead to incorrect taxonomic assignment.

Part 8 — assignTaxonomy

What assignTaxonomy does:

This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

Taxonomy assignment to ASV visually reflects the species and community composition from samples, as well as their roles.

Why might some ASVs not be assigned all the way to species?

Some ASVs may not be assigned all the way to species due to sequence resolution, reference database limitations, or primarily sequenced regions being too short.

Part 9 — Sequence counts summary

What sequence counts shows:

This output tracks how many reads remain after each major step in the pipeline so you can see where reads were retained or lost.

Why is it useful to track reads across steps?

TOLD TO SKIP PART 9 BY VL

Upload screenshot: sequence counts output

Choose File no file selected

Approximately how many reads did you start with?

NA

Approximately how many reads remained at the end?

NA

At which step did you lose the most reads?

NA

What is one likely explanation for that loss?

NA

Part 10 — Moving into phyloseq

After completing DADA2 in Galaxy, we can create a phyloseq object for analysis.

10A. Which files will you take from DADA2 from Galaxy to use in phyloseq?

ASV/sequence table, taxonomy table, metadata file.

10B. Why must sample names in the metadata match sample names in the ASV table?

Consistent formatting for sample names across metadata and the ASV table is essential for correctly linking the data.

10C. In 2–3 sentences, explain how DADA2 transforms raw reads into data that can be analyzed ecologically in phyloseq.

Dada2 tools allow us to visualize read quality and trim sequences accordingly, determine error rates for denoising, assess accuracy through merging sequence pairs, remove chimeras, and assign taxonomy to ASVs. Overall, assessing and increasing sequence accuracy improves downstream analysis in phyloseq.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset

Project/sample set: Bird microbiome

Sequencing type: paired-end Illumina

Grouping variable: infection status

Quality profiles

Purpose: The quality profile tool aids in visualizing and understanding the distribution of respective quality scores.

Reverse vs forward quality: Reverse reads are typically lower quality than forward reads because by the time the first read has been sequenced and the second can start, the signal has degraded and accuracy declines.

Forward quality drop: around 220

Reverse quality drop: around 200 bp

Forward truncation: NA

Reverse truncation: NA

Why these values: I chose truncation values a margin away from those values whereby quality began to drop in order to keep as much overlap as possible for merging. Willing to tolerate maybe a quality score above 25- up to you.

filterAndTrim

Purpose: Trimming from the ends and filtering enhances the accuracy of downstream processing by helping to eliminate sequencing errors.

Parameters used:

```
"Truncate read length" was set to 220; selected yes to "separate filters and trimmers for reverse reads;" "trim start of each read" was set to 10; "truncate read length" was set to 205.
```

If trimmed too much: Poor merging and loss of usable data can result from overtrimming, reducing coverage/depth of sequence(s).

If not trimmed enough: Low quality sequences can increase error rates.

learnErrors

Purpose: The learnErrors step is important because it determines how and how often errors occur.

Model fit: The fitted model mostly follows the observed error pattern.

Why important: Learning error rates is important to perform before denoising because it allows dada2 to distinguish between true sequence variation and sequencing errors.

dada + mergePairs

Denoising purpose: When provided filtererd reads and error reads, it uses the learned error model to correct errors and reconstruct true sequences (amplicon sequence variants).

ASVs vs OTUs: OTU clustering groups of sequences into taxa based on similarity threshold without modeling errors, whereas ASV Inference corrects sequencing errors at the nucleotide level. ASV provides more precision in VL's opinion.

mergePairs purpose: Forward and reverse reads are being merged to assess confidence of sequence accuracy.

Need for overlap: Overlap between forward and reverse reads confirms that they match and indicates accuracy.

One reason reads fail to merge: Reads may fail to merge if there isn't enough overlap.

Sequence table + chimera removal + taxonomy

Sequence table purpose: The ASV table represents diversity with precise comparison of sampled sequences. there are about 1600 ASVs.

Rows/columns/cells: Columns are ASVs, samples, and frequency of ASV appearance in each sample.

Chimera removal purpose: Chimeras are sequences artificially generated during PCR and because they

don't actually exist in the original sample the presence of chimeras compromises sequence accuracy and downstream processing.

Why chimeras are a problem: The presence of chimeras can lead to incorrect taxonomic assignment.

Taxonomy purpose: Taxonomy assignment to ASV visually reflects the species and community composition from samples, as well as their roles.

Taxonomy limits: Some ASVs may not be assigned all the way to species due to sequence resolution, reference database limitations, or primarily sequenced regions being too short.

Sequence counts

Purpose: TOLD TO SKIP PART 9 BY VL

Starting reads: NA

Ending reads: NA

Largest loss step: NA

Likely explanation: NA

Move to phyloseq

Files to download:

ASV/sequence table, taxonomy table, metadata file.

Why names must match: Consistent formatting for sample names across metadata and the ASV table is essential for correctly linking the data.

DADA2 to phyloseq transition: Dada2 tools allow us to visualize read quality and trim sequences accordingly, determine error rates for denoising, assess accuracy through merging sequence pairs, remove chimeras, and assign taxonomy to ASVs. Overall, assessing and increasing sequence accuracy improves downstream analysis in phyloseq.